

SE LAB 1

Problem Statement 48

Incident Escalation & On-Call Rotation Engine

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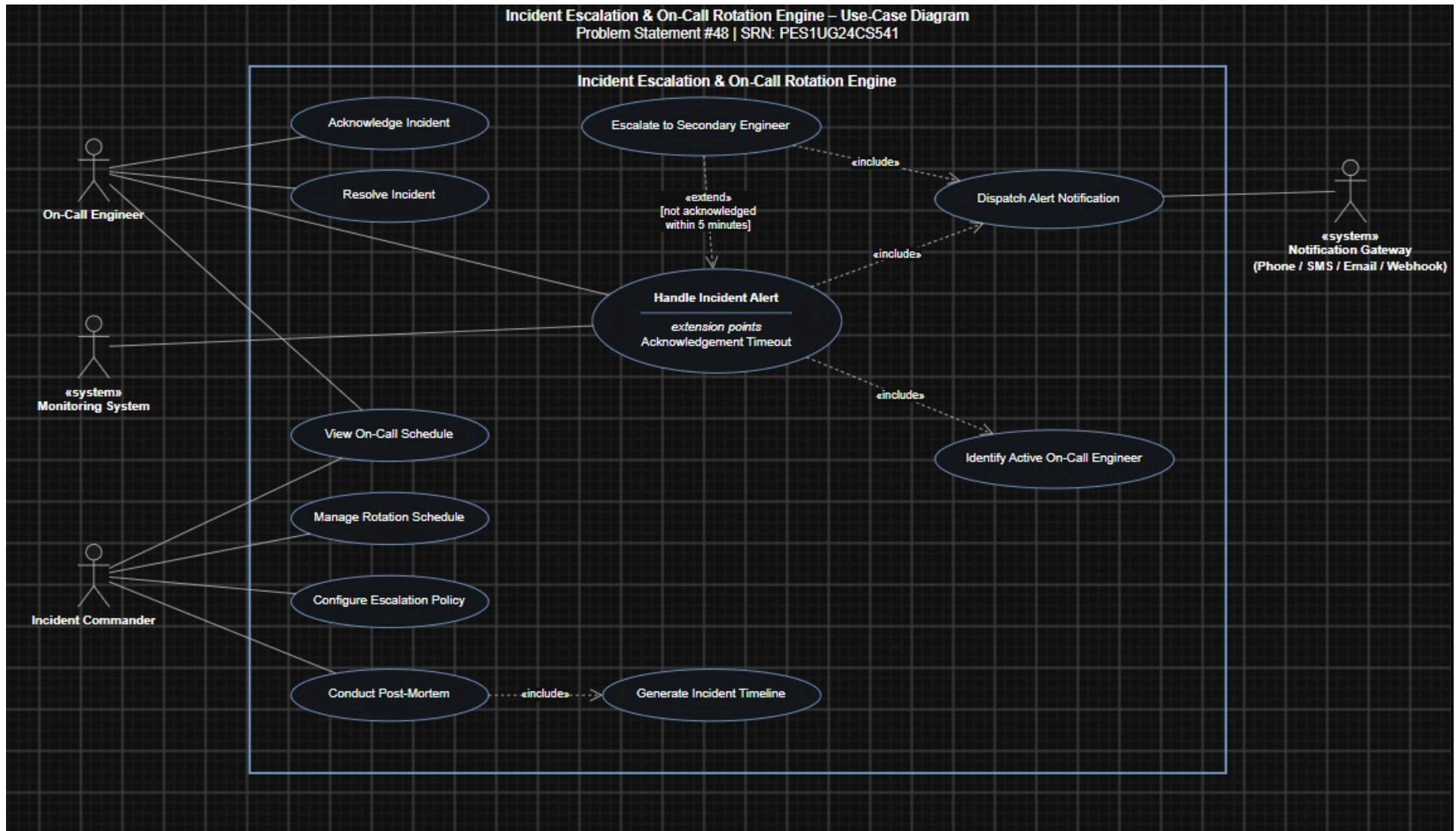
Section: I

Requirements Table

Problem Statement 48: Incident Escalation & On-Call Rotation Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest incident alerts, identify the active on-call engineer from the rotation schedule, and escalate to secondary engineer if unacknowledged within 5 minutes.	High	Pass: Escalation step triggered upon timeout; the secondary engineer is notified and the escalation is recorded in the incident log. Fail: Unacknowledged P1 incident drops without escalation.	Ensures no critical incident goes unattended when the primary engineer is unavailable.
FR-002	Functional	The system shall allow the Incident Commander to create, edit, and publish team rotation schedules with a primary and secondary on-call engineer per shift, including temporary overrides for shift swaps.	High	Pass: Every timestamp in the schedule resolves to exactly one primary and one secondary engineer, and a schedule with a coverage gap is blocked from publishing. Fail: A timestamp resolves to no engineer or conflicting engineers.	Provides the accurate, gap-free rotation that alert routing depends on.
FR-003	Functional	The system shall allow the Incident Commander to configure a multi-tier escalation policy (Primary Engineer, Secondary Engineer, Incident Commander) with per-tier channels (phone, SMS, email, webhook) and a timeout of 1 to 5 minutes.	High	Pass: An unacknowledged test P1 alert is notified tier by tier on the configured channels and reaches the Incident Commander; a timeout above 5 minutes is rejected. Fail: A tier or channel is skipped or the ladder stops early.	Implements the multi-tiered phone/SMS escalation ladder consistently with FR-001.
FR-004	Functional	The system shall allow the On-Call Engineer to acknowledge an incident by SMS reply, phone keypress, or web console and later resolve it, cancelling any pending escalation and timestamping every state change.	High	Pass: An acknowledgement before the timeout moves the incident to Acknowledged, no further escalation notifications are sent, and all state changes are logged. Fail: Escalation continues after a valid acknowledgement or a change is not logged.	Stops needless paging and establishes clear incident ownership.
FR-005	Functional	The system shall generate an incident timeline when a P1 or P2 incident is resolved and allow the Incident Commander to record a post-mortem with root cause and owned action items before closing the incident.	Medium	Pass: A complete, chronologically ordered timeline is produced and the incident closes only after a post-mortem with at least one owned action item is submitted. Fail: Timeline events are missing or the incident closes without a post-mortem.	Turns incidents into documented learning that prevents recurrence.
NFR-001	Nonfunctional (Performance & Security)	Alert dispatch via webhook, SMS, and email must initiate within 3 seconds of alert ingestion.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load: at 1,000 alerts per minute the 99th-percentile dispatch initiation time is 3 seconds or less, all traffic uses TLS 1.2+, webhooks are HMAC-signed, and unauthenticated alerts are rejected. Fail: Latency exceeds 3 seconds or a security check fails.	Every second of delay extends downtime, and alert endpoints must not be spoofable.
NFR-002	Nonfunctional (Reliability & Availability)	The alert ingestion and escalation services shall maintain at least 99.95% monthly availability, lose no accepted alert, persist pending escalation timers, and fail over to a standby node within 60 seconds.	High	Pass: A fault-injection test that kills the active node during pending escalations shows failover within 60 seconds, zero lost alerts, and every escalation firing within 60 seconds of schedule. Fail: An alert or escalation is lost or failover exceeds 60 seconds.	The on-call engine must stay up while other systems are failing.

UML Use-Case Diagram



Use-Case Flow Specification

Use Case	UC-01 - Handle Incident Alert
Primary Actor	On-Call Engineer
Supporting Actors	Monitoring System; Notification Gateway; Incident Commander
Trigger	The Monitoring System sends an incident alert to the engine's ingestion endpoint.

Preconditions

- A published, gap-free rotation schedule assigns a primary and a secondary on-call engineer for the affected service.
- An escalation policy (Primary Engineer, Secondary Engineer, Incident Commander) with a 5-minute acknowledgement timeout is configured, and the Monitoring System holds valid API credentials.

Postconditions

- The incident is Acknowledged and owned by an engineer, pending escalation timers are cancelled, and every step is logged with timestamps.
- If no one acknowledges, the incident keeps escalating along the ladder and is never silently dropped.

Main Success Scenario

1. The Monitoring System sends a P1 incident alert to the engine.
2. The system authenticates the request, validates the payload, and creates an incident in the Triggered state.
3. The system identifies the active primary and secondary on-call engineers from the rotation schedule.
4. The system dispatches the alert to the primary On-Call Engineer via phone, SMS, email, and webhook within 3 seconds of ingestion.
5. The system starts the 5-minute acknowledgement timer.
6. The primary On-Call Engineer acknowledges the incident before the 5-minute timer expires.
7. The system moves the incident to Acknowledged, assigns ownership to the engineer, and cancels the pending escalation timer.
8. The system notifies the incident channel of the acknowledgement and records every step in the incident log.
9. The On-Call Engineer begins investigation, continuing in the Resolve Incident use case.

Alternate Flow - 6a: Incident Not Acknowledged Within 5 Minutes

- The 5-minute timer expires, so the system marks the incident Escalated, logs the escalation, and dispatches the alert to the secondary On-Call Engineer with a new 5-minute timer.
- If the secondary engineer also does not acknowledge within 5 minutes, the system escalates to the Incident Commander through all channels.
- When any tier acknowledges, the system cancels the remaining timers and the flow resumes at Step 7; the incident is never dropped without escalation.